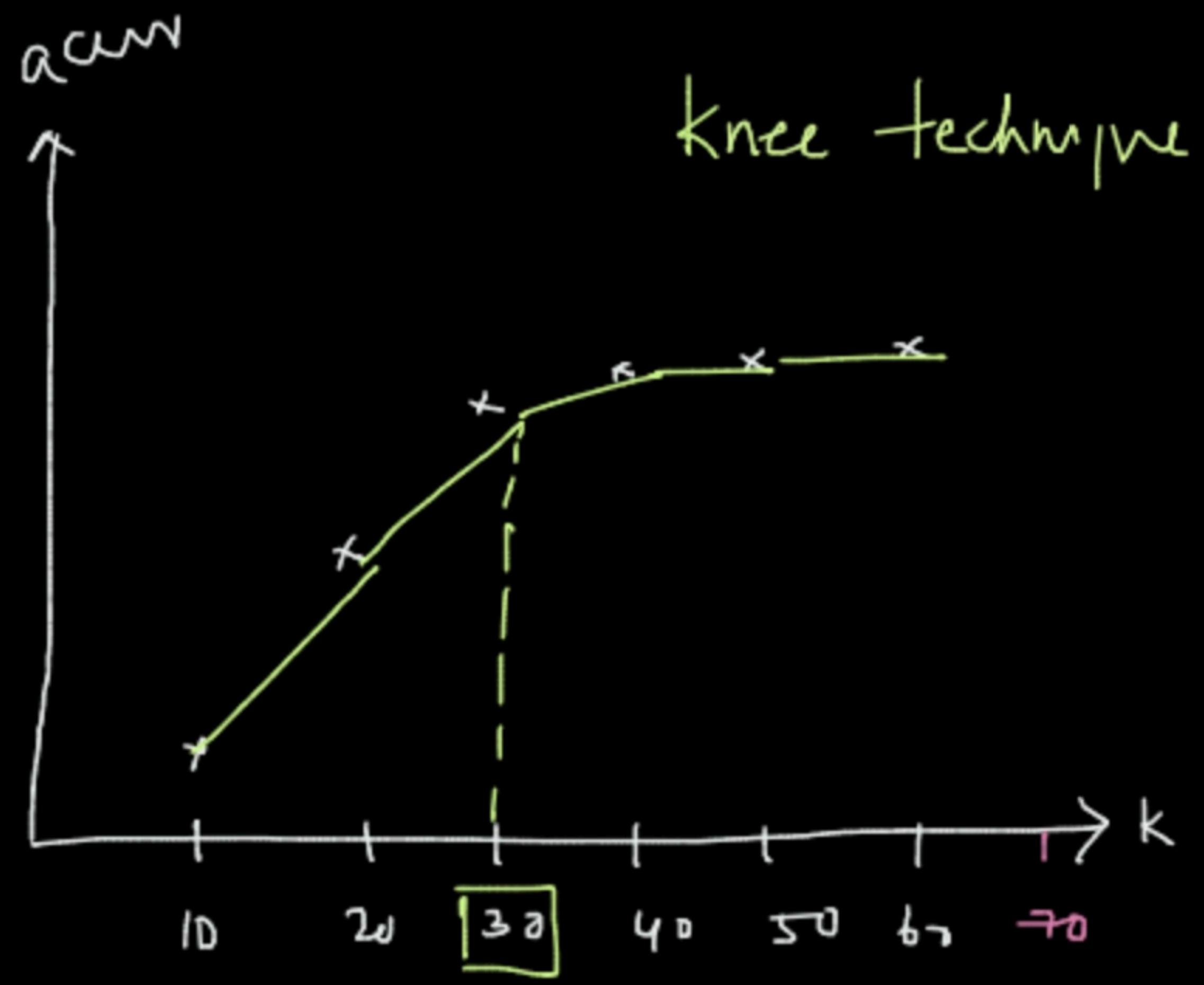
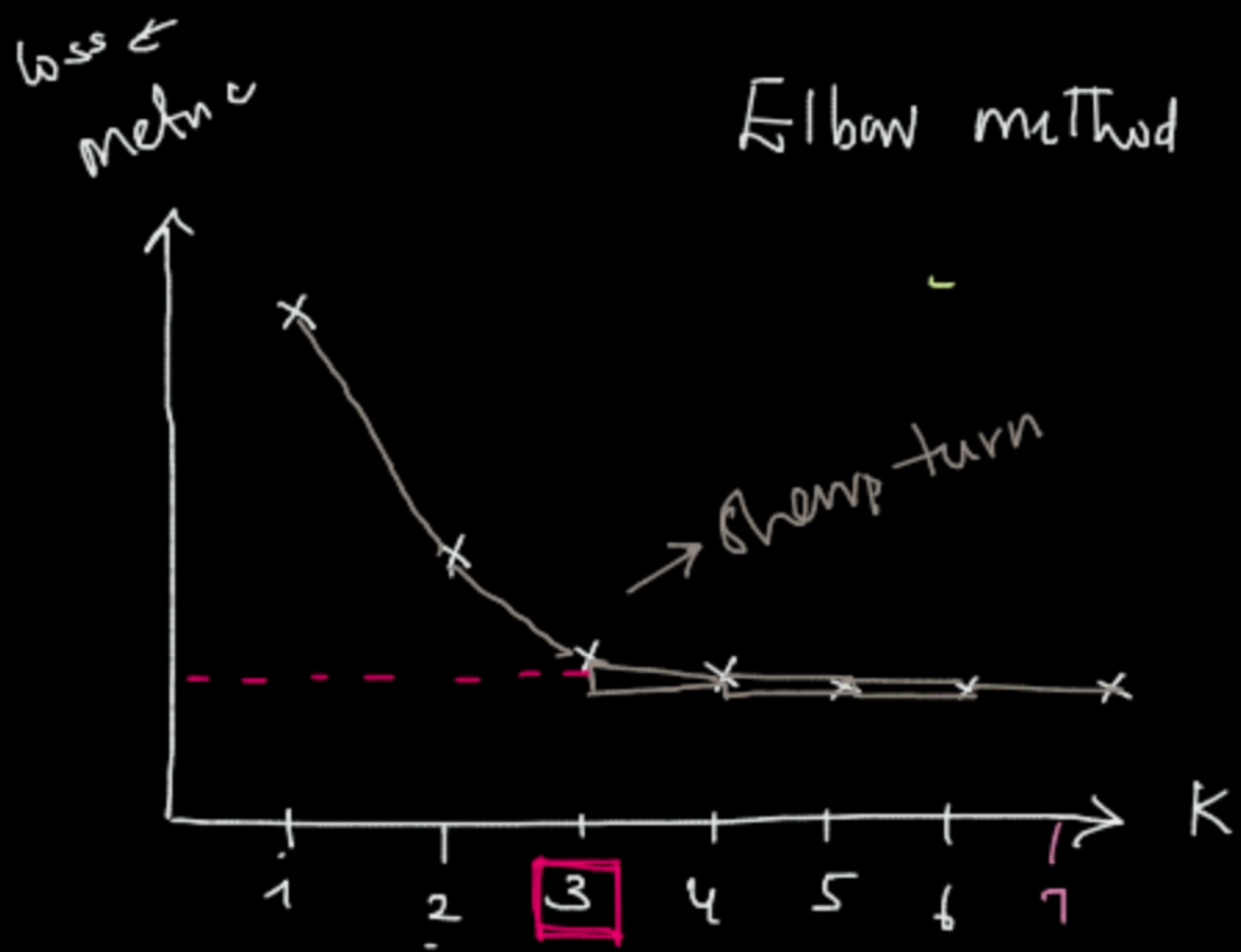




Hyperparameter tuning

only one hp $\rightarrow$ k





$\{ k_1 : [1 \ 2 \ 3 \ 4 \ 5]$
 $k_2 : [10 \ 20 \ 30 \ 40 \ 50]$
 $k_3 : [0.1 \ 0.2 \ 0.3 \ 0.4] \}$

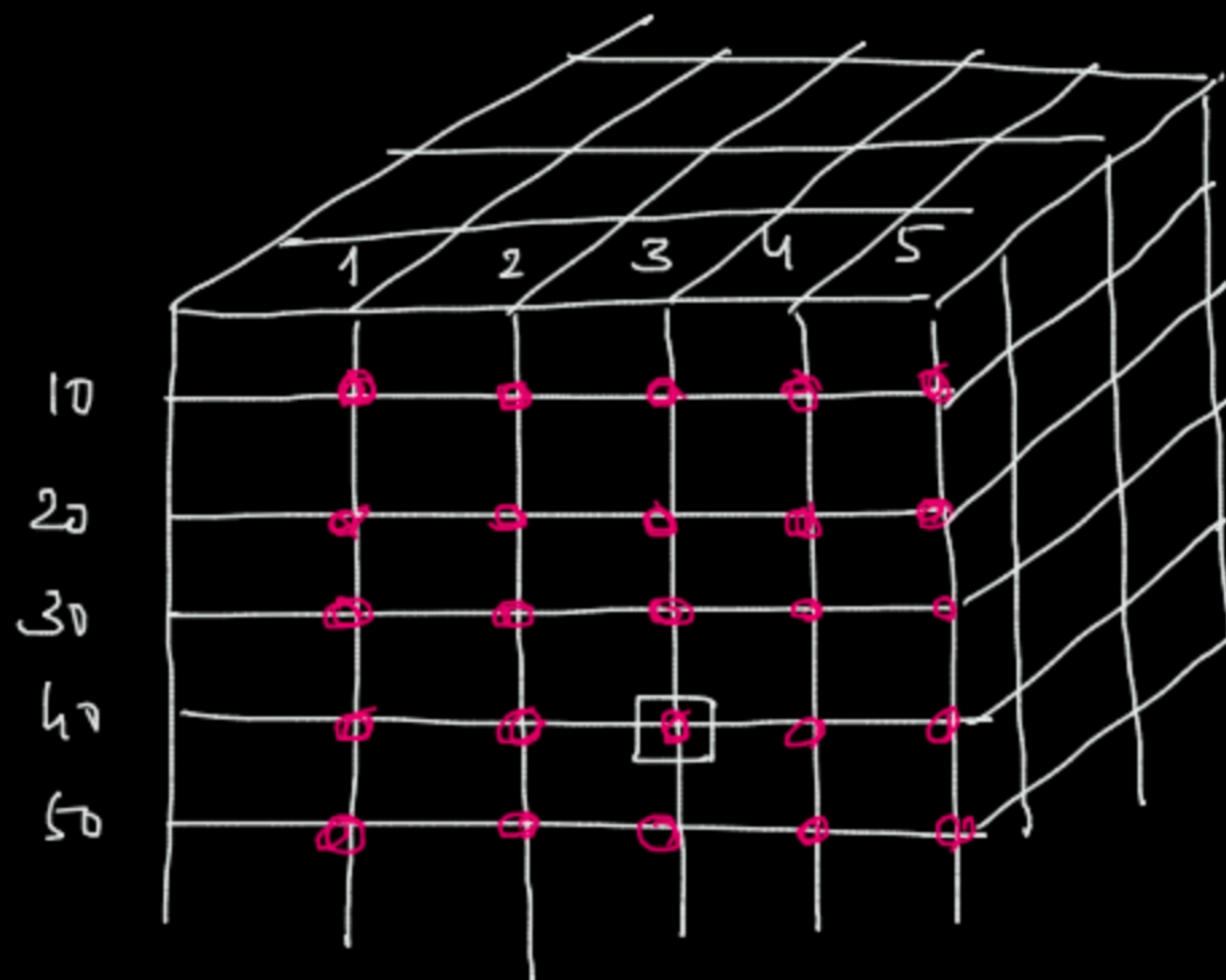
$$4 \times 5 \times 5 = 25 \times 4 = 100$$

Grid search $\rightarrow \checkmark$

C:

gamma:

Kernel: 'rbf', 'poly', 'linear'
 $\uparrow$



$\rightarrow k_1 = 3 \quad k_2 = 40$
 $\downarrow$





Dimensionality Reduction

— Reducing no of features without losing information

— pre-processing stage

Why? → 'Curse of dimensionality'

↳ models don't work well in high dimensions

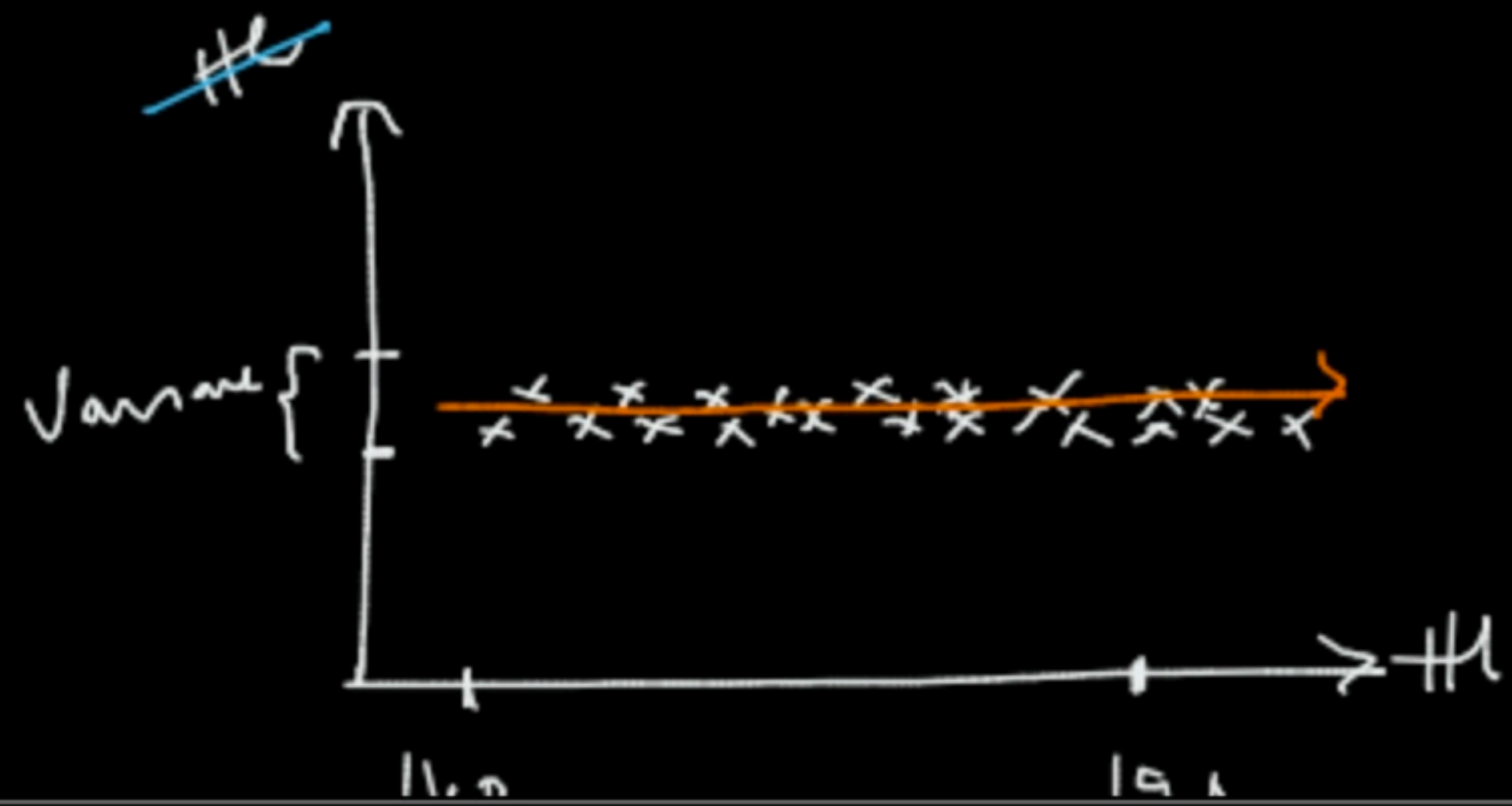
Model Building ←

→ Space & time Complexity is high (PCA)

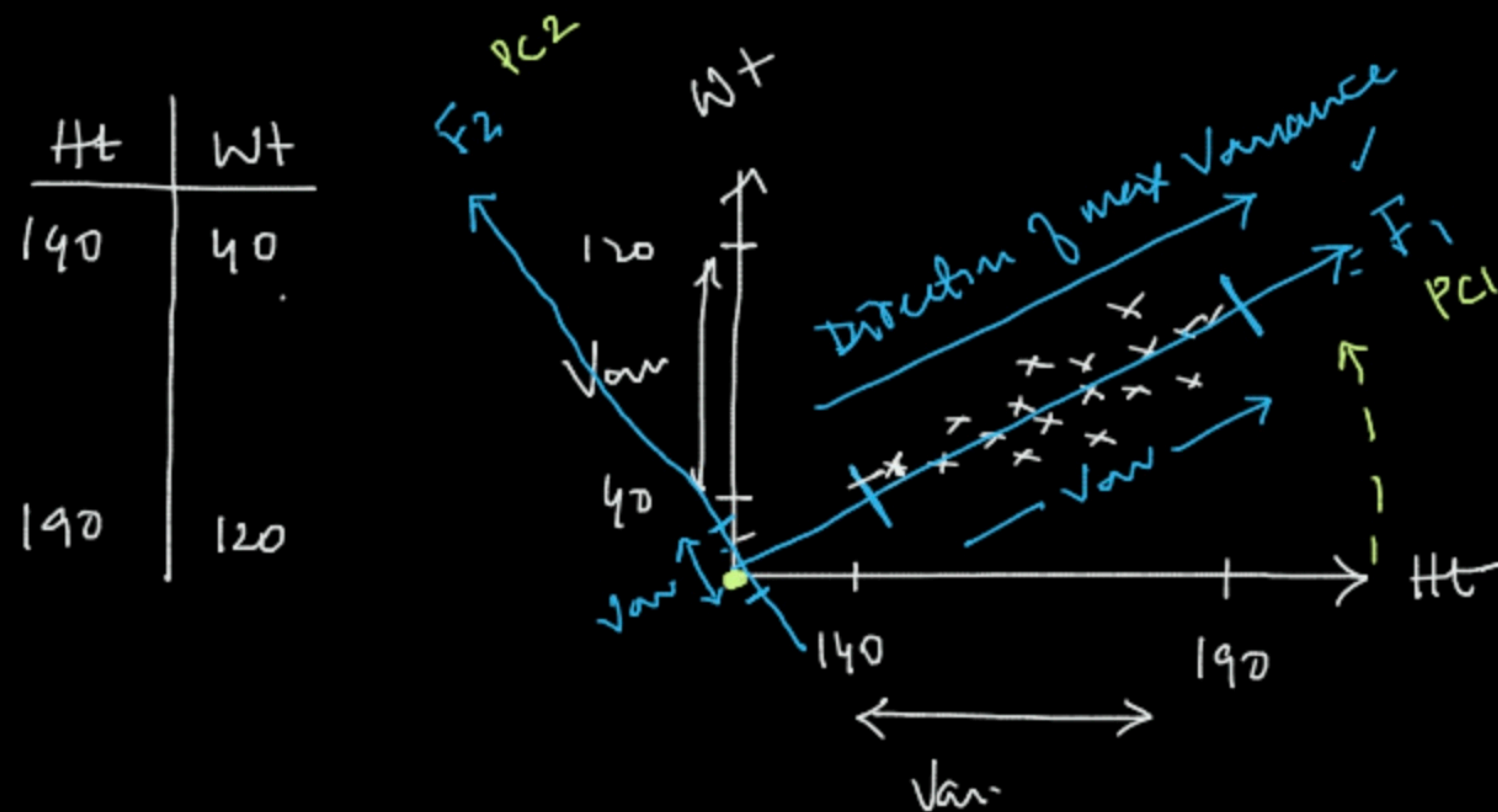
Visualizing data → only 2d can be visualised

(TSNE) → T-distributed stochastic neighbourhood embedding

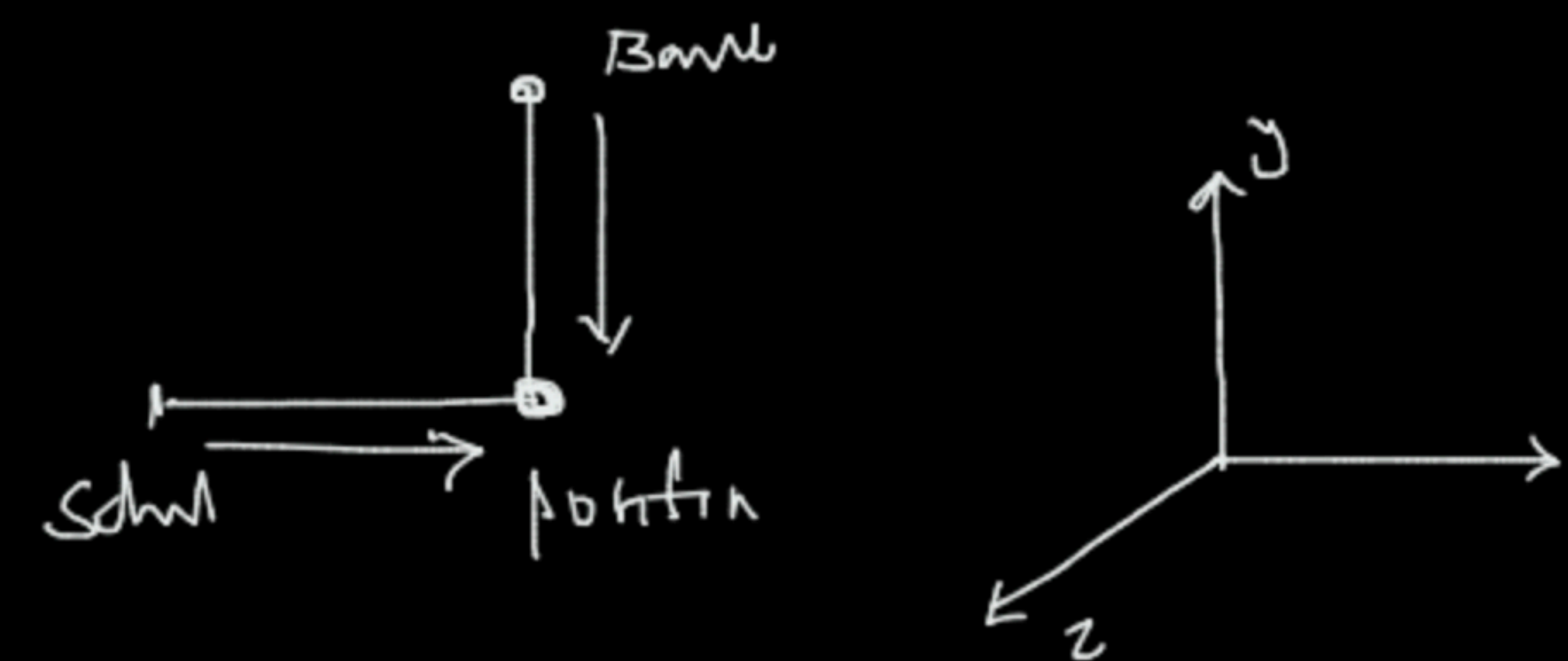
Principal Component Analysis → PCA



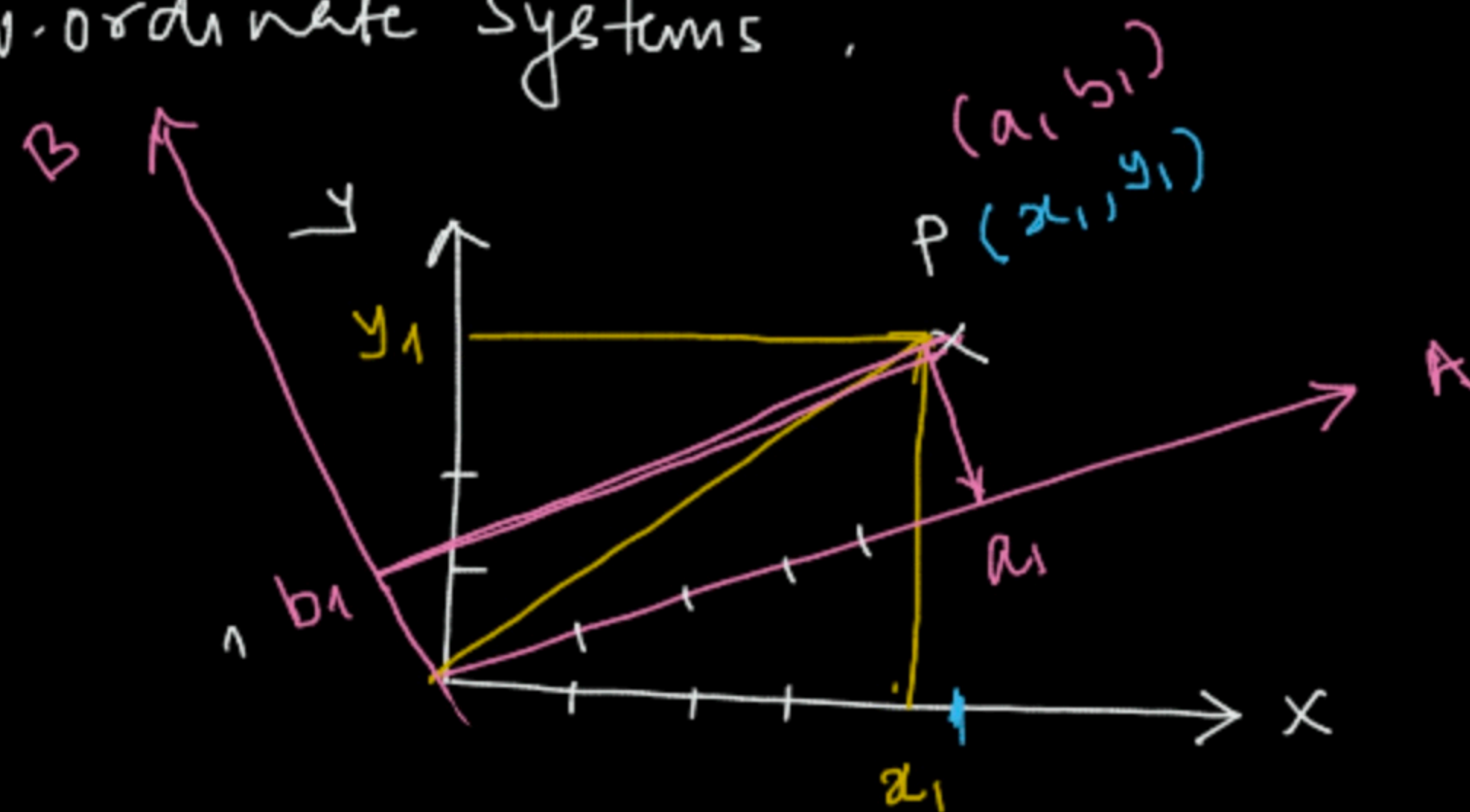
	H1	H2
P1	140	0.9
P2		1.0
P3		0.87
P4	140	0.95



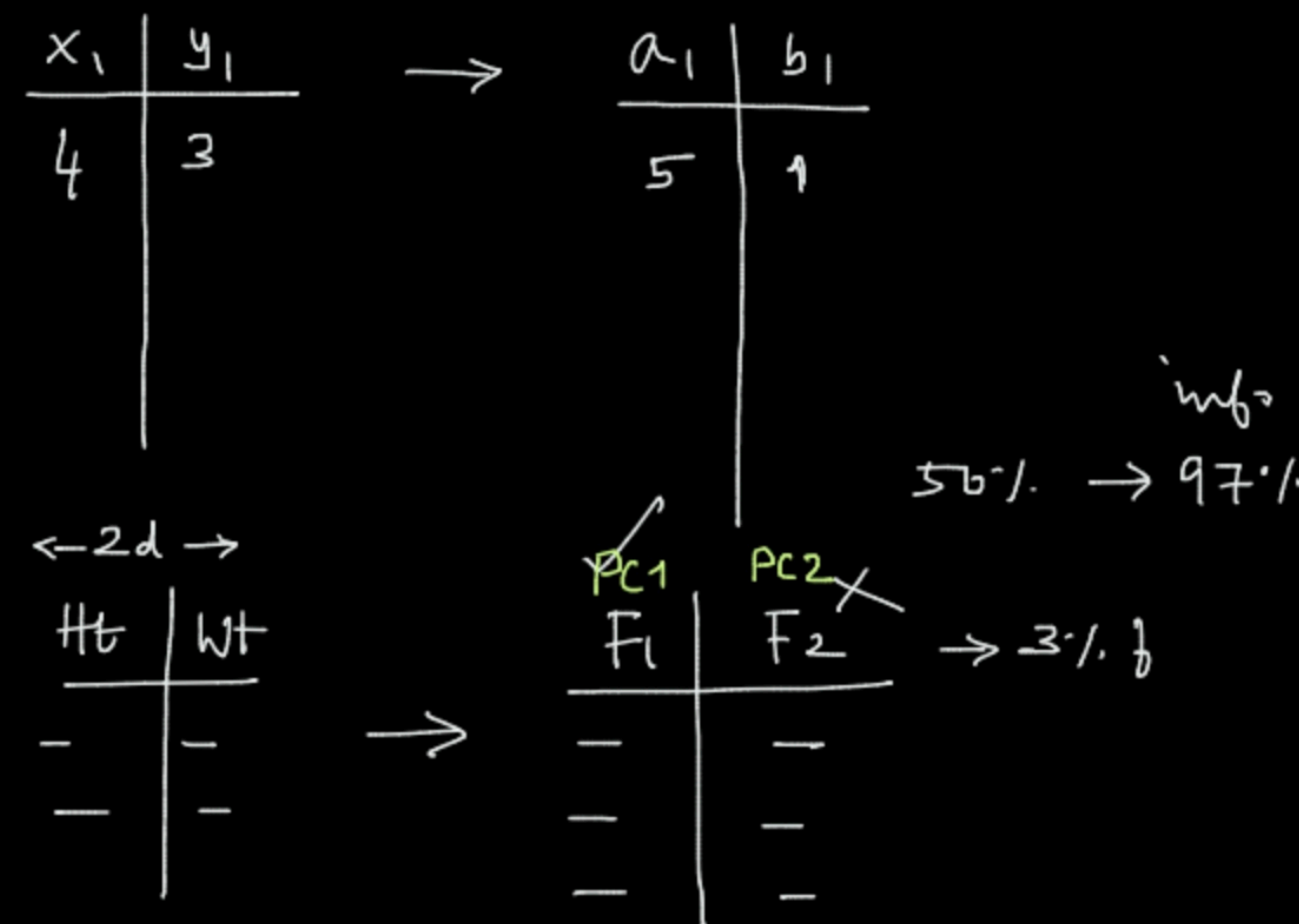
"Without losing information"



Coordinate systems:



Orthogonal Transformation $\rightarrow$



Step 1: Standardize the data

→ 48

$$CV(x, x) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

= Variance Variance

Correlation $\rightarrow \frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y}$

$$V_1 \perp V_2 \perp V_3$$

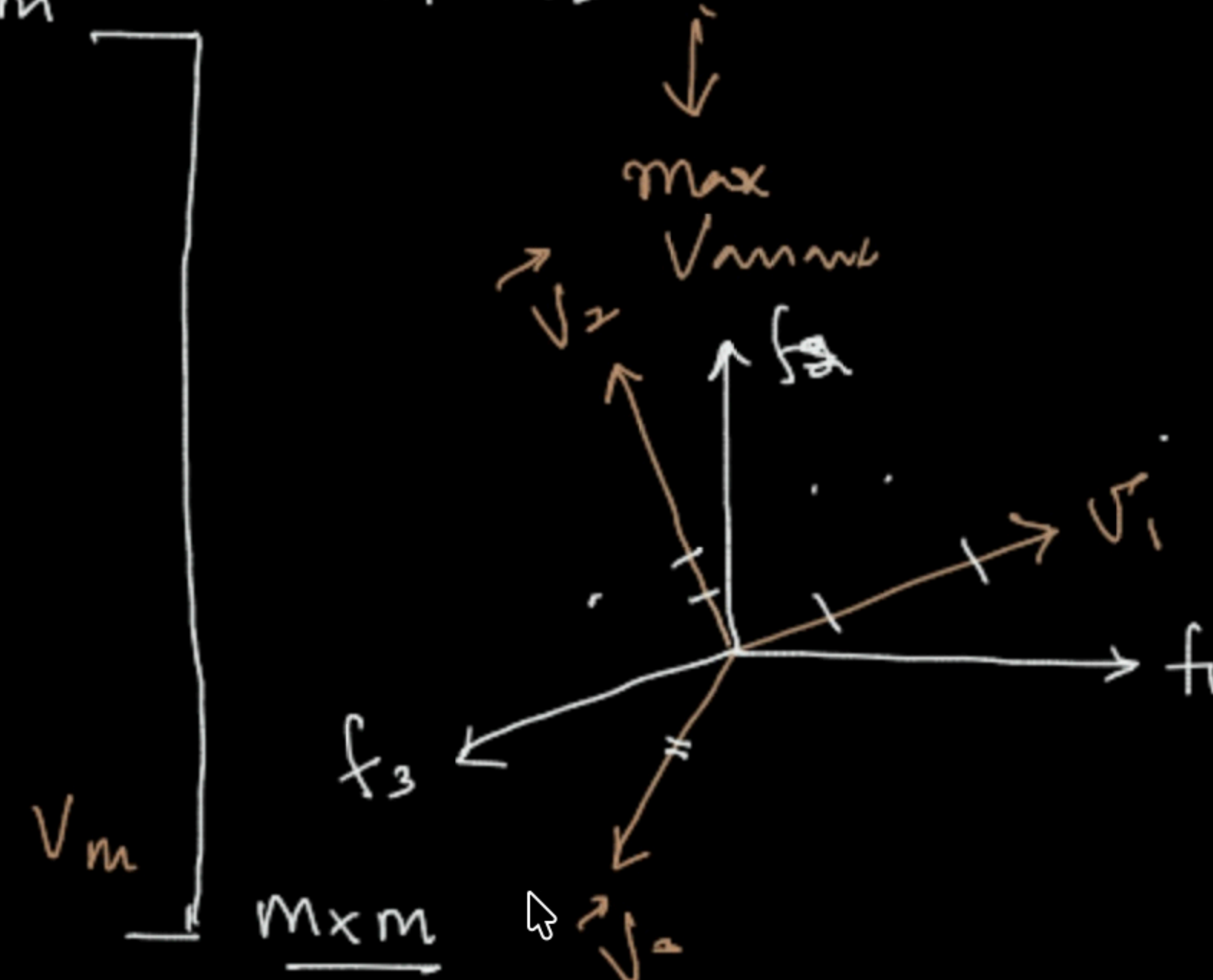
$$\underbrace{\lambda_1 \quad \lambda_2 \quad \lambda_3 \quad \dots \quad \lambda_m}_{\text{variance}}$$

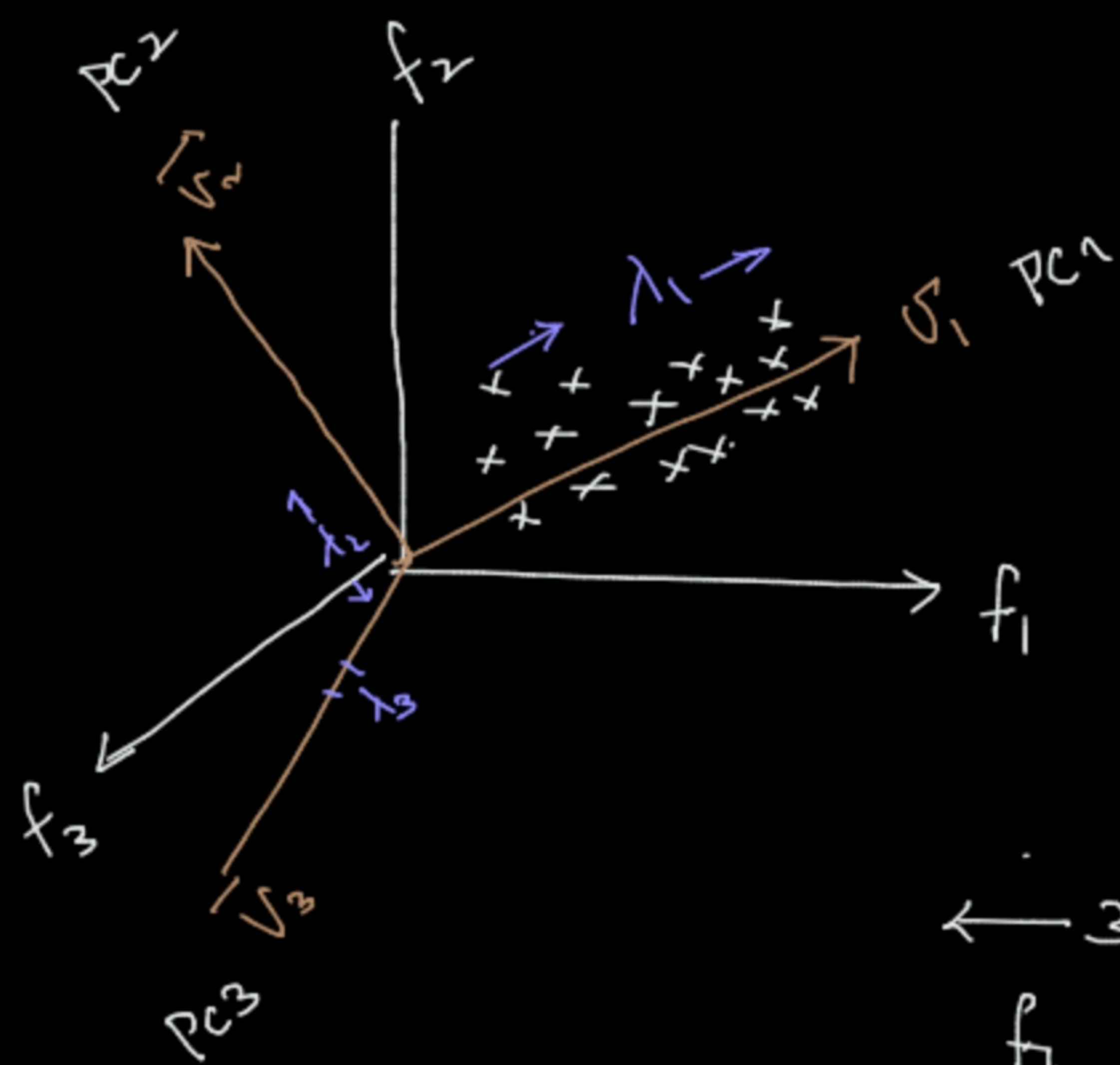
$$\underbrace{\vec{v}_1 \quad \vec{v}_2 \quad \vec{v}_3 \quad \dots \quad \vec{v}_m}_{\text{eigenvectors}}$$

	f_1	f_2	f_3	$\dots$	f_m
1	—	—	—		—
2	—	—	—		—
3	—				
$\vdots$					
$\vdots$					
$\vdots$					
n					

$$C_N(x)$$

$$\begin{array}{c|cccc}
 & f_1 & f_2 & f_3 & \dots & f_m \\
 \hline
 f_1 & v_1 & & & & \\
 f_2 & c_{21} & v_2 & & & \\
 f_3 & c_{31} & c_{32} & v_3 & & \\
 \vdots & & & & & \\
 f_m & f_{m1} & & & & v_m
 \end{array}$$





$$\vec{v}_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} \quad 85\%$$

$$\vec{v}_2 = \frac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} \quad 10\%$$

$$\vec{v}_3 = \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} \quad 5\%$$

$$PC1 \rightarrow a_1 f_1 + a_2 f_2 + a_3 f_3$$

← 3d →

f_1	f_2	f_3
-	-	-
-	-	-
-	-	-

PC1	PC2	PC3
-	-	-
-	-	-
-	-	-

→ Eigen Value Decomposition ✓

SVD →

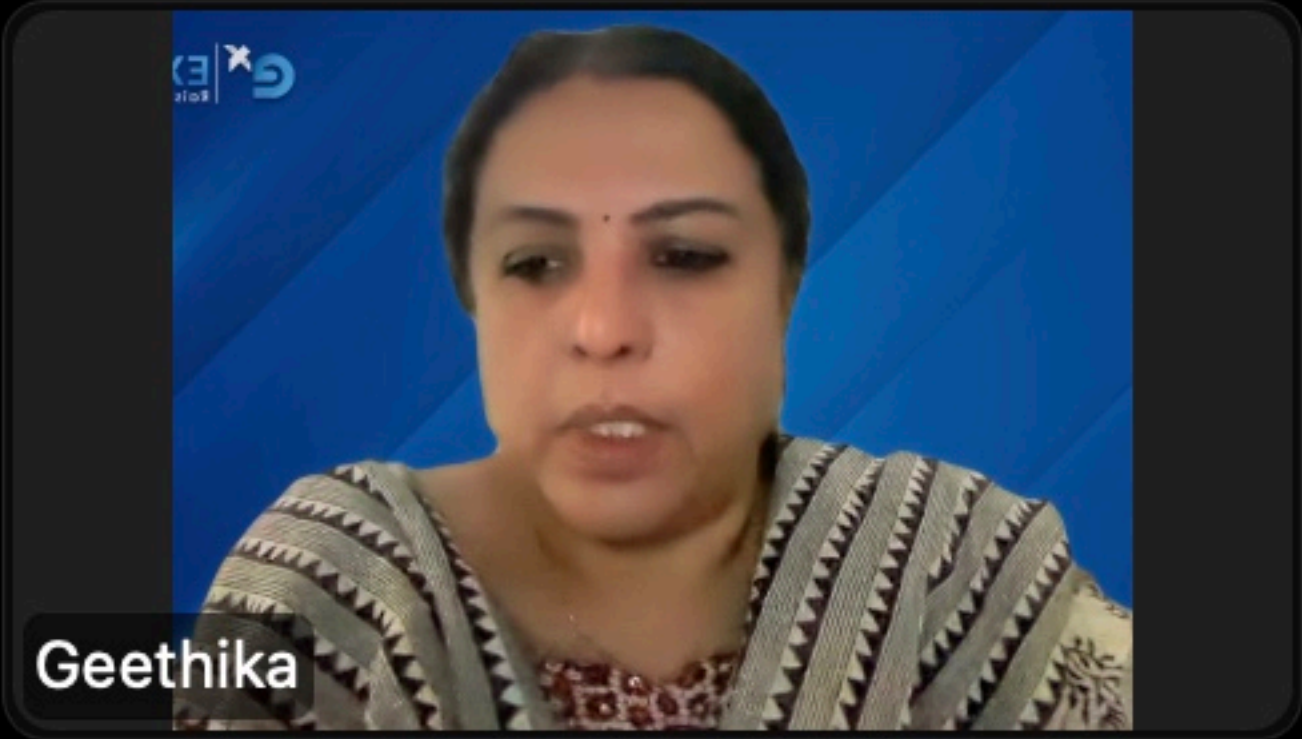
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